

MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Computer Science and Engineering

Faculty of Innovation Engineering

Final Year Project Proposal

Title: Lightweight Hand Gesture Recognition and Eye Tracking for Offline Interaction on AI Glasses

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Abstract

This project proposes an offline visual interaction system for an AI glasses prototype, combining first-person hand gesture recognition with eye tracking on a resource-constrained embedded platform. An outward-facing camera will capture hand movements, while an inward-facing camera will capture the wearer's eye region. The system will recognize a small set of practical gestures and estimate eye opening, blinking, and coarse gaze direction. Calibrated gaze-based target selection will be investigated as an extension.

The main research focus is the relationship between recognition quality, interaction latency, model size, and power consumption. MediaPipe Hand Landmarker will provide a baseline and candidate landmark labels for training a lightweight hand model. Eye-related supervision will be obtained through reviewed annotations and calibration sequences, with the suitability of existing landmark tools assessed separately. Compact student models will be developed and evaluated using supervised learning, output-based knowledge distillation where compatible teacher outputs are available, and INT8 quantization.

The embedded pipeline will combine region-of-interest tracking, bounded frame queues, temporal filtering, and adaptive inference scheduling. A state machine will translate visual predictions into interaction events such as selection, clicking, dragging, and cancellation. All core functions will operate without an Internet connection.

The expected outputs are a working prototype, a reproducible data and training pipeline, independently deployable lightweight models, and comparative experiments on recognition accuracy, latency, memory usage, power consumption, and continuous-operation stability.

Contributions

The individual contributions to the preparation of this proposal will be recorded according to the actual work completed by each member.

Member	Contribution to the proposal
[Member 1]	[Describe the background research, requirements analysis, or technical planning actually completed.]
[Member 2, if applicable]	[Describe the actual contribution.]
[Member 3, if applicable]	[Describe the actual contribution.]

These descriptions will be updated before submission to ensure that they accurately represent each member's independent work.

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1. Introduction

AI glasses provide a platform for visual interaction while allowing users to keep their hands available for other activities. However, interaction is constrained by the size, weight, processing capacity, and battery capacity of wearable hardware. A useful interaction system must provide reliable input without introducing excessive delay or energy consumption.

Hand gestures can express commands such as selection, cancellation, and dragging. In a first-person configuration, an outward-facing camera observes the wearer's hand from a moving viewpoint. Changes in illumination, background, hand orientation, occlusion, and motion blur make recognition challenging. Although an existing recognition framework can establish a functional baseline, developing a deployable prototype also requires model training, hardware integration, and measurement of actual device performance.

Eye tracking offers a complementary input. Gaze can indicate a potential target, while a deliberate hand gesture can confirm the intended action. Separating target selection from confirmation may help prevent actions caused by ordinary looking behaviour. The initial eye-related functions will be eye-region tracking, blink detection, and coarse gaze estimation. Precise gaze-point estimation will depend on camera geometry, suitable supervision, and user calibration.

A forward-facing camera cannot directly observe the wearer's eyes. Therefore, the proposed prototype requires a separate inward-facing imaging path. During development, the computing board and battery may be carried separately from the glasses frame, while the cameras remain mounted near the wearer's head. This arrangement will allow the project to investigate the algorithms and runtime behaviour before attempting further physical miniaturization.

The project will study three questions:

1. How much model size and inference cost can be reduced while preserving useful gesture-recognition performance?
2. How do detection frequency, tracking, and adaptive scheduling affect interaction latency and measured power consumption?
3. Can eye-related input and hand gestures be combined into reliable interaction events within the available hardware resources?

The expected contribution is an implemented and experimentally evaluated system that demonstrates the trade-offs involved in offline wearable interaction.

2. Objectives

The overall objective is to develop an AI glasses prototype that performs visual interaction locally on an embedded platform.

The specific objectives are:

1. Establish the camera and hardware pipeline. Capture first-person hand video and near-eye video, associate frames with timestamps, and verify stable acquisition on the selected platform.

2. Develop practical gesture recognition. Detect and track a single hand and recognize open palm, fist, pointing, and pinch gestures. Interpret pinch-and-release, repeated pinches, and pinch-and-move sequences as click, double-click, and drag events.

3. Implement eye-related perception. Track the visible eye region and estimate eye opening, blinking, and coarse gaze direction. Investigate calibrated target selection after the basic eye pipeline is reliable.

4. Train independently deployable lightweight models. Build student models using reviewed labels and teacher-generated supervision where appropriate. Compare a supervised baseline with a distilled model under consistent experimental conditions.

5. Reduce computation and energy use. Evaluate compact network structures, INT8 quantization, region-of-interest processing, detector-tracker separation, and adaptive inference frequency.

6. Integrate reliable interaction logic. Use confidence thresholds, temporal filtering, hysteresis, and event-state transitions to reduce false or repeated actions.

7. Evaluate the complete system on real hardware. Measure recognition quality, end-to-end latency, memory usage, power consumption, temperature, and stability during continuous operation.

The minimum deliverable will include an offline hand-interaction pipeline, basic eye tracking and blink detection, a trained lightweight model, and device-based comparative experiments. Calibrated gaze selection and NPU acceleration will be developed after the core pipeline is established.

3. Related work

3.1 Hand Detection and Tracking

MediaPipe Hands uses a palm detector and a hand landmark model to estimate the hand skeleton from RGB images [Zhang2020]. The current Hand Landmarker documentation also describes a tracking mechanism that can avoid repeated palm detection when tracking remains successful [GoogleHand].

These approaches provide a useful reference for separating detection from tracking. In this project, MediaPipe will support baseline evaluation and candidate label generation. Teacher predictions will be reviewed because errors caused by occlusion or unusual viewpoints could otherwise be transferred to the student model.

3.2 Lightweight Neural Networks

MobileNetV3 combines network design with hardware-aware optimization and includes a Small variant intended for lower-resource applications [Howard2019]. It provides a candidate backbone for the proposed student models.

The project will assess input resolution, channel width, and output-head complexity on the selected device. Architectural compactness alone will not be treated as proof of lower system power consumption; device measurements will be required.

3.3 Knowledge Distillation

Hinton, Vinyals, and Dean describe knowledge distillation as a way to transfer information from a larger model into a smaller model [Hinton2015]. The proposed project will distinguish teacher-generated pseudo-label training from distillation using compatible teacher outputs.

For hand landmarks, the student may learn reviewed teacher coordinates. For gesture classification, a teacher classifier can be constructed from landmark features if suitable probability outputs are available. The value of this supervision will be tested against training with the same manually reviewed labels alone.

3.4 Quantization

Jacob et al. investigate quantization and training methods for integer-only neural-network inference [Jacob2018]. This work motivates the evaluation of INT8 deployment in the proposed system.

Post-training quantization will be attempted first using representative calibration data. If quantization causes substantial task-specific degradation, quantization-aware training will be considered. Accuracy, latency, model size, and measured power will be reported separately.

3.5 Eye Tracking and Gaze Estimation

Zhang et al. study appearance-based gaze estimation under varying real-world conditions [Zhang2015]. Their work highlights the importance of evaluating generalization across users and imaging conditions.

MediaPipe Face Landmarker produces facial landmarks and expression-related outputs through a face-detection pipeline [GoogleFace]. Its suitability for close-up eye images cannot be assumed, and facial landmarks are not equivalent to calibrated gaze measurements. Consequently, the proposed near-eye pipeline will use independently verified eye annotations and known fixation targets rather than relying exclusively on Face Landmarker outputs.

4. Methodology

4.1 System Architecture

The system will contain a development pipeline and an embedded runtime pipeline.

The development pipeline will perform data preparation, teacher inference, label review, student-model training, quantization, and evaluation. The embedded pipeline will perform camera acquisition, preprocessing, local inference, temporal filtering, and event generation.

The runtime flow will be:

Camera acquisition → Latest-frame buffer → Detection or tracking → Region-of-interest processing → Lightweight model inference → Temporal filtering → Interaction state machine → Application event

The hand and eye modules will initially be implemented separately. Integration will occur after their outputs and failure behaviour have been validated.

4.2 Hardware Feasibility

The first stage will verify camera mounting, image quality, simultaneous acquisition, and available processing capacity. The inward-facing arrangement will be designed around the eye region that can be observed reliably. Tracking both eyes will depend on the final field of view and camera arrangement.

The final platform will be selected after these tests. An ARM-based board will provide the CPU baseline. NPU acceleration will be evaluated only after model compatibility, camera bandwidth, power requirements, and runtime support have been checked.

4.3 Data Collection and Annotation

A pilot dataset will be collected before larger-scale recording. Hand recordings will cover different gestures, backgrounds, lighting conditions, hand distances, and movement speeds. Eye recordings will include open and closed eyes, blinks, and fixation on known target positions.

Participants will receive an explanation of the collection purpose and data handling procedure before recording. Participant identifiers will be anonymized, and raw recordings will remain in controlled local storage.

MediaPipe Hand Landmarker will generate candidate hand labels. Invalid landmarks, low-quality tracking intervals, and abrupt coordinate changes will be reviewed or marked unusable. Gesture classes and dynamic events will be checked against the recorded actions.

Eye opening and blink labels will be manually reviewed. Gaze supervision will be obtained from fixation sequences with known target positions and documented camera geometry. An instruction to look at a target will not automatically establish a precise three-dimensional gaze direction unless the required geometry is available.

Training, validation, and test sets will be separated by participant. An additional repeated-session test will assess whether user calibration remains useful after the device is removed and worn again. Adjacent frames from the same recording will not be distributed randomly across training and test sets.

4.4 Hand Gesture Recognition

The hand pipeline will begin with low-resolution detection. A tracker will maintain the hand region between detector executions, and the detector will run again when tracking quality falls below a threshold.

A compact image model or landmark-based model will be developed for the selected static gestures. Landmark coordinates, where used, will be normalized to reduce sensitivity to hand position and scale.

Dynamic interactions will be defined through explicit temporal states. A click will require a valid pinch followed by release. A double-click will require two completed click sequences within a defined interval. Dragging will require a maintained pinch and movement, followed by release. Low-confidence predictions and missing-hand states will suppress new actions.

4.5 Eye Tracking and Multimodal Interaction

The eye pipeline will estimate eye opening and blinking from a cropped eye region. Coarse gaze estimation will initially use a limited set of target directions or regions. More precise gaze mapping will be investigated through a short calibration procedure.

The demonstrator will use a controlled interface with known target locations. A gaze estimate will identify a candidate target, and a pinch will confirm selection. The system will require a sufficiently stable target estimate before accepting confirmation.

If calibrated gaze selection is unreliable, the prototype will retain a gesture-controlled interface and report eye-tracking outputs separately. This fallback will preserve a demonstrable core system while making the limitations of gaze interaction explicit.

4.6 Training and Model Compression

A floating-point student baseline will be trained first. Candidate architectures will use compact convolutional blocks and task-specific output heads.

Experiments will compare supervised training with additional teacher supervision. Distillation losses will be used only when teacher and student outputs have compatible meanings. Reviewed gesture labels and calibration-derived eye labels will remain necessary to avoid treating teacher predictions as unquestioned ground truth.

The selected models will then undergo INT8 quantization. Structured channel reduction or pruning may be investigated if supported by the deployment runtime and development schedule.

4.7 Runtime Scheduling

The runtime will have standby, tracking, and interaction states. Standby will use low-frequency presence checks. Tracking will maintain regions of interest with periodic detection. Interaction will increase processing frequency when a deliberate action is detected.

State transitions will include hysteresis to avoid rapid switching. Frame buffers will have a fixed capacity, and stale frames will be discarded to prevent growing interaction delay. Model files and required dependencies will be installed locally before deployment.

4.8 Evaluation

Hand evaluation will include macro F1, confusion matrices, event precision and recall, and false or repeated trigger rates. Eye evaluation will include blink F1, tracking availability, recovery time, and coarse gaze-region accuracy. Calibrated gaze error will be reported only when a suitable reference measurement is available.

System evaluation will measure:

- End-to-end latency, including P50, P95, and P99.
- Processing rate and frame-interval variation.
- Model size and memory usage.

• Average and peak device power.

• Temperature and performance during continuous operation.

Comparisons will include the baseline, floating-point student, distilled student where applicable, and INT8 student. Fixed-frequency and adaptive scheduling will use the same workload. Energy per valid interaction will be reported together with recognition quality so that missed events do not create a misleading efficiency advantage.

An initial continuous-operation test will last at least 30 minutes. Final acceptance thresholds will be established after pilot measurements and documented before the final experiments.

5. Required hardware and software

5.1 Hardware

Component	Intended purpose
ARM-based development board, with Raspberry Pi 5 as a candidate	Camera acquisition, preprocessing, CPU inference, and event logic
Outward-facing wide-angle RGB camera	First-person hand capture
Inward-facing compact camera or camera assembly	Near-eye image capture
Optional compatible NPU accelerator	Evaluation of accelerated quantized inference
microSD card or equivalent storage	Operating system, models, and logs
Regulated power supply and portable battery	Development and wearable demonstration
Power measurement instrument	Voltage, current, power, and energy measurement
Cooling solution	Continuous-operation temperature management
Adjustable head-mounted camera frame	Repeatable positioning
Computer or monitor	Controlled interaction demonstration and data inspection

Final models and quantities will be confirmed after feasibility testing. The first prototype may use a separately carried computing board and battery.

5.2 Software

The proposed software includes Python and PyTorch for training, OpenCV for image processing, MediaPipe for baseline inference and candidate annotations, and Git for version control. Data analysis and plotting will use standard scientific Python tools.

An embedded inference runtime supporting the selected CPU and model format will be chosen after deployment tests. If an NPU is used, its supported conversion tools and runtime will be evaluated separately. The final runtime will include all required models and libraries for offline operation.

6. Project planning

Development will run from October 2026 to April 2027. Writing and experiment documentation will continue throughout implementation.

The preliminary Gantt chart below shows planned activity periods. Shaded cells indicate active work.

Task	Oct 2026	Nov	Dec	Jan 2027	Feb	Mar	Apr
Literature review and proposal	■	■					
Hardware and camera feasibility	■	■					
Pilot data and annotation pipeline		■	■				
Baseline and student-model training		■	■	■			
Eye tracking and calibration		■	■	■	■		
Quantization and embedded deployment			■	■	■		
Interaction integration and scheduling				■	■	■	
Comparative evaluation					■	■	■
Report writing and revision	■	■	■	■	■	■	■
Demonstration and defense preparation						■	■

The submission milestones are:

Date	Deliverable
1 November 2026	Project proposal
30 November 2026	Progress report
7 December 2026	Oral progress presentation
12 April 2027	Final report PDF for defense
19-23 April 2027	Defense and project demonstration
26 April 2027	Revised final report PDF

The principal risks are near-eye image quality, insufficient teacher-label quality, dual-camera processing cost, quantization degradation, and thermal instability. Camera feasibility will therefore be tested early. Manual label review will supplement automatic annotations, and the CPU baseline will be retained if accelerator integration is unsuccessful. Basic eye tracking and gesture interaction will take priority over precise gaze selection.

The expected final deliverables are the prototype, data preparation tools, trained models, an offline deployment package, experiment records, the final report, and a working demonstration.

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